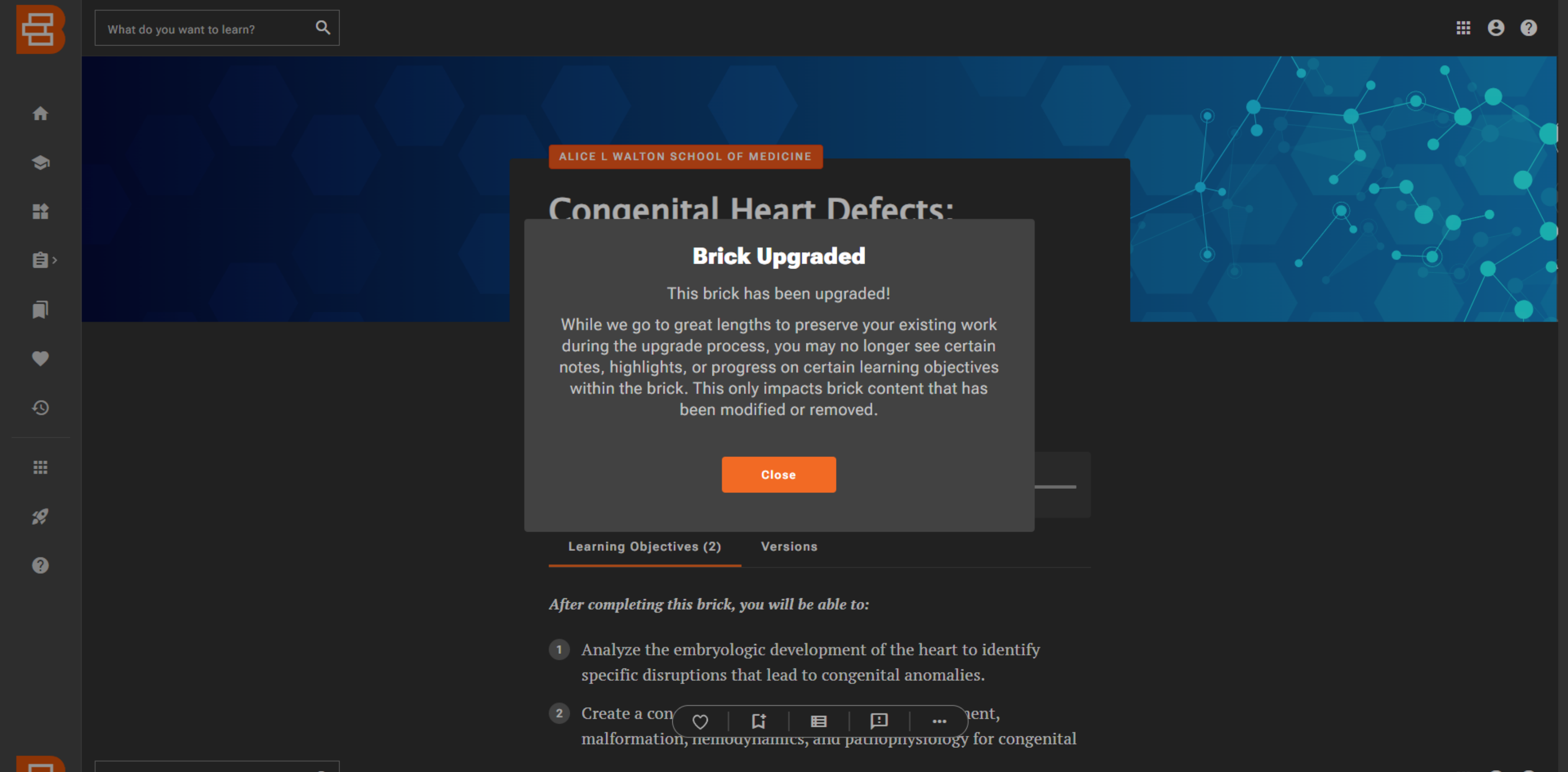


Congenital Heart Defects:

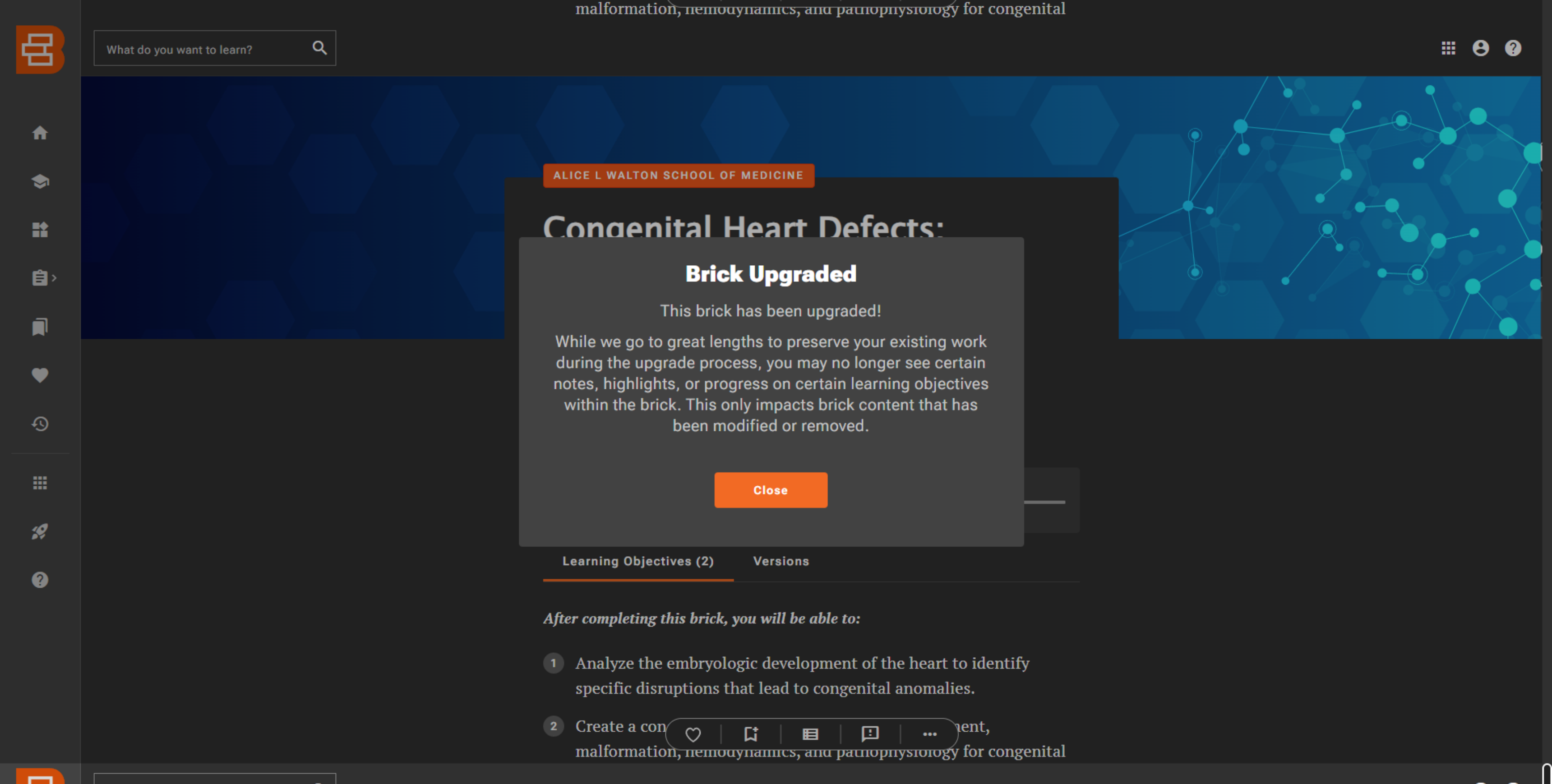
Brick Upgraded

This brick has been upgraded!

While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.

After completing this brick, you will be able to:

1. Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2. Create a con ent, malformation, hemodynamics, and pathophysiology for congenital


Congenital Heart Defects:

Brick Upgraded

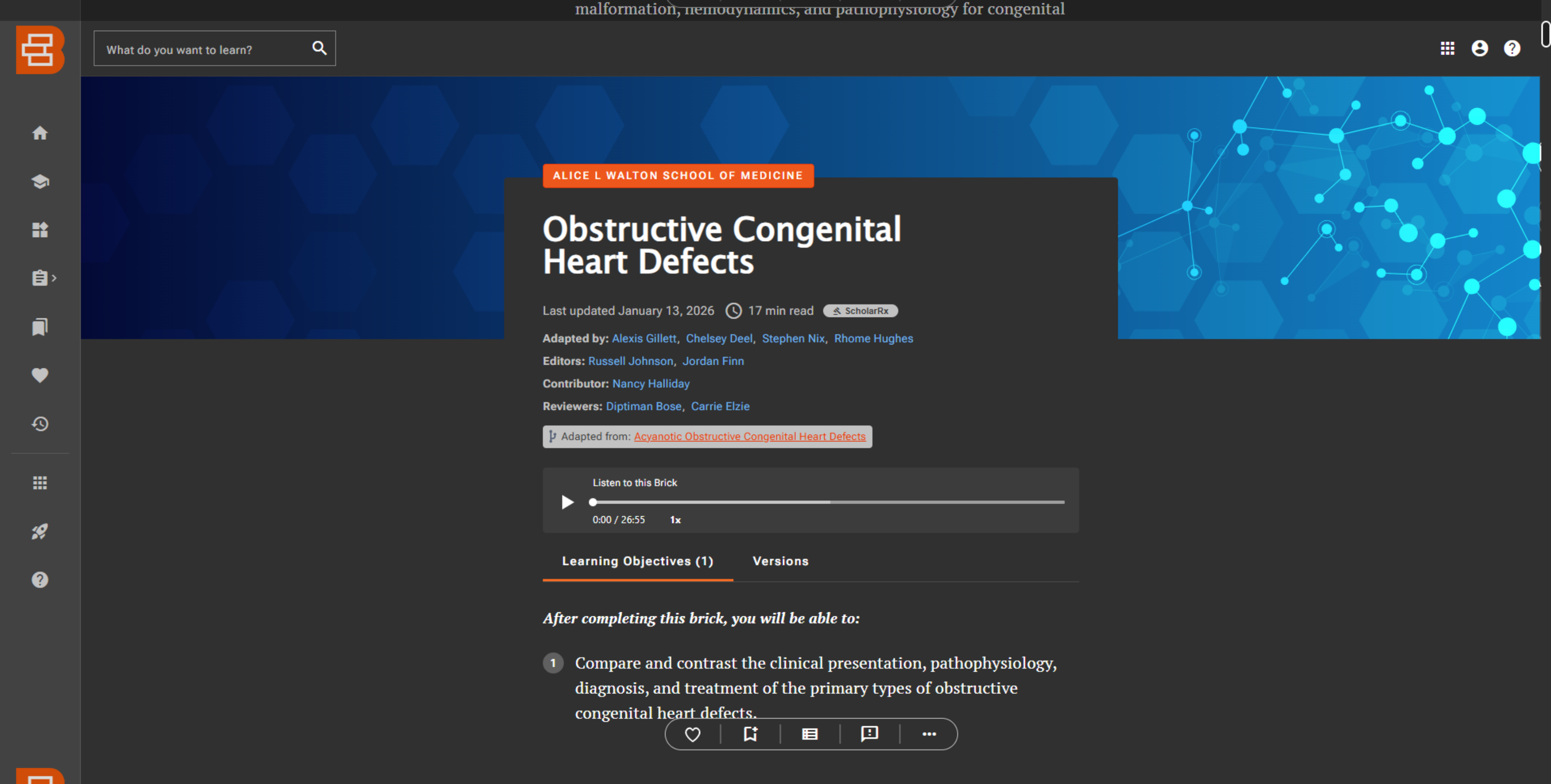
This brick has been upgraded!

While we go to great lengths to preserve your existing work during the upgrade process, you may no longer see certain notes, highlights, or progress on certain learning objectives within the brick. This only impacts brick content that has been modified or removed.

After completing this brick, you will be able to:

1. Analyze the embryologic development of the heart to identify specific disruptions that lead to congenital anomalies.
2. Create a con ent, malformation, hemodynamics, and pathophysiology for congenital

---



Obstructive Congenital Heart Defects

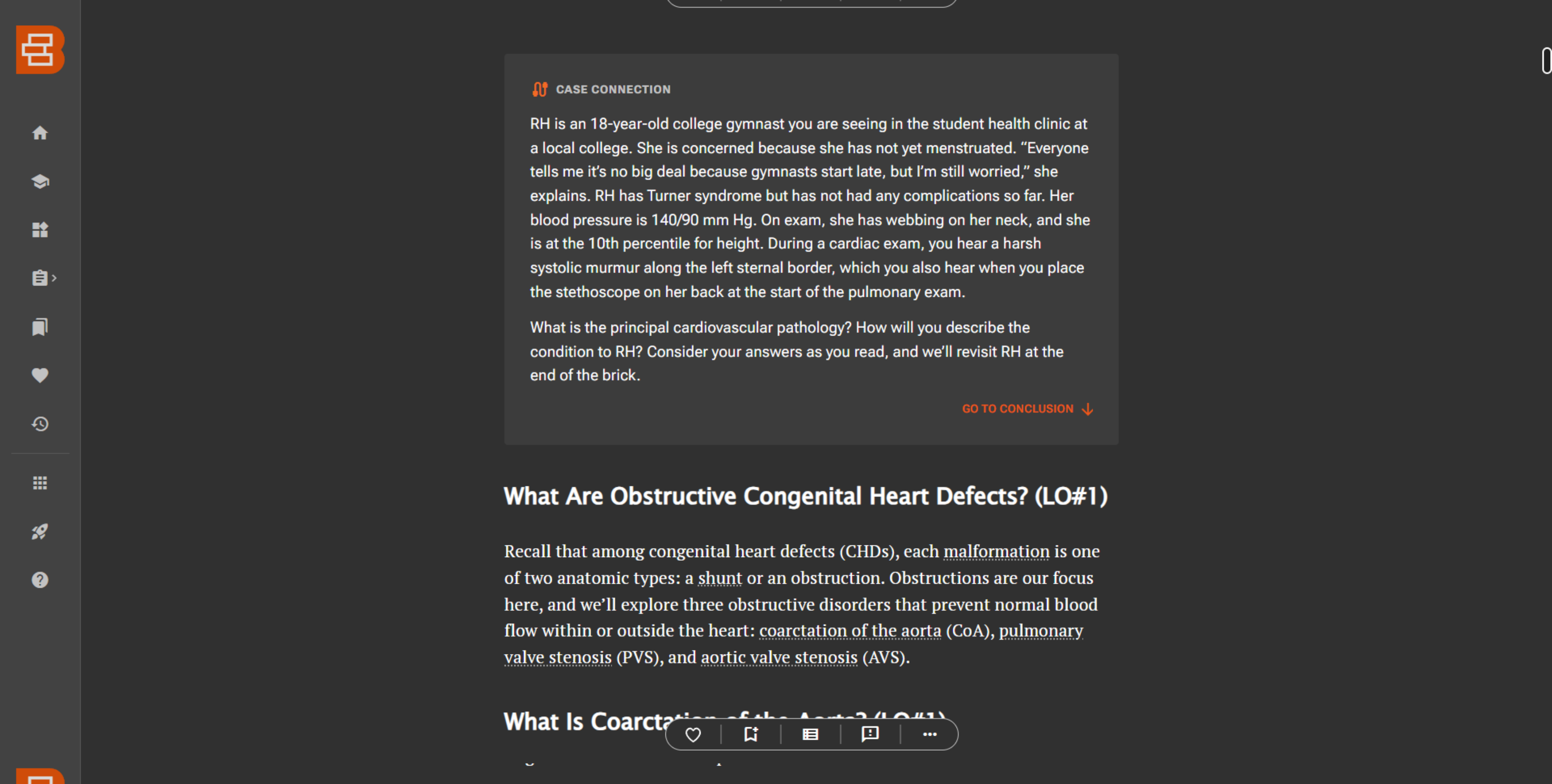
Last updated January 13, 2026 17 min read

Adapted by: Alexis Gillett, Chelsey Deel, Stephen Nix, Rhome Hughes
Editors: Russell Johnson, Jordan Finn
Contributor: Nancy Halliday
Reviewers: Diptiman Bose, Carrie Elzie

Adapted from: Acyanotic Obstructive Congenital Heart Defects

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of obstructive congenital heart defects.



CASE CONNECTION

RH is an 18-year-old college gymnast you are seeing in the student health clinic at a local college. She is concerned because she has not yet menstruated. “Everyone tells me it’s no big deal because gymnasts start late, but I’m still worried,” she explains. RH has Turner syndrome but has not had any complications so far. Her blood pressure is 140/90 mm Hg. On exam, she has webbing on her neck, and she is at the 10th percentile for height. During a cardiac exam, you hear a harsh systolic murmur along the left sternal border, which you also hear when you place the stethoscope on her back at the start of the pulmonary exam.

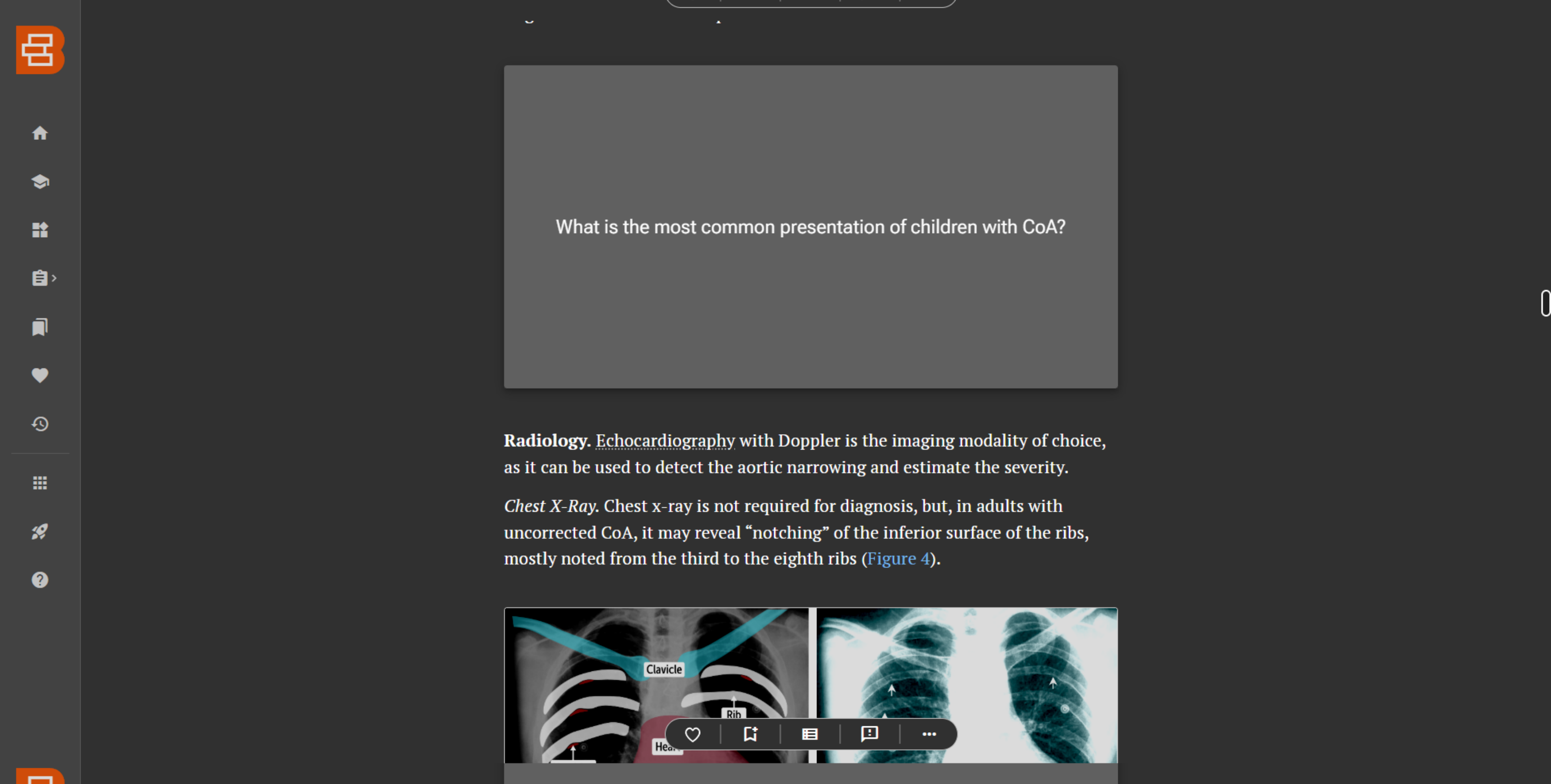
What is the principal cardiovascular pathology? How will you describe the condition to RH? Consider your answers as you read, and we’ll revisit RH at the end of the brick.

GO TO CONCLUSION ↓

What Are Obstructive Congenital Heart Defects? (LO#1)

Recall that among congenital heart defects (CHDs), each malformation is one of two anatomic types: a shunt or an obstruction. Obstructions are our focus here, and we’ll explore three obstructive disorders that prevent normal blood flow within or outside the heart: coarctation of the aorta (CoA), pulmonary valve stenosis (PVS), and aortic valve stenosis (AVS).

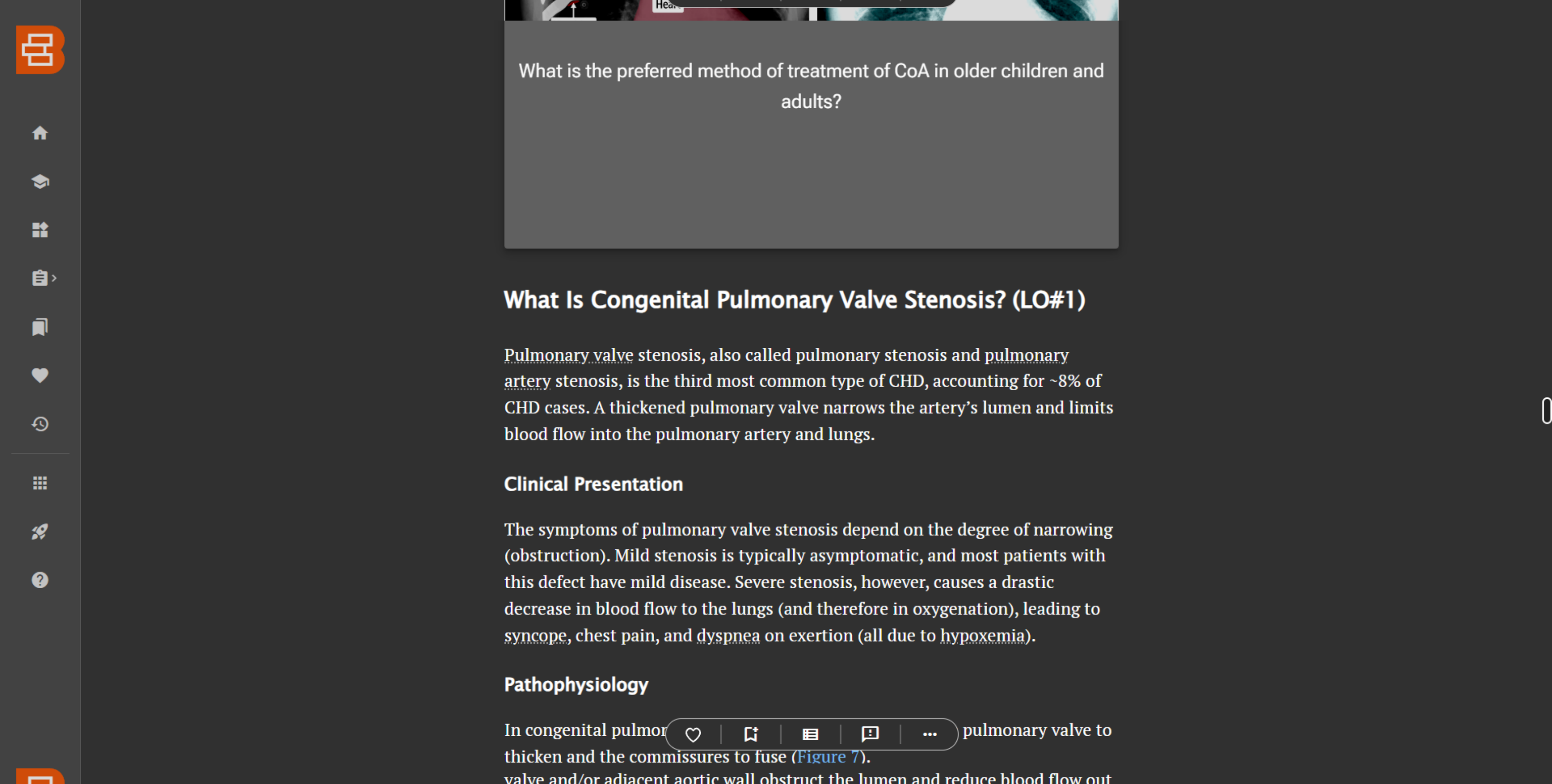
What Is Coarctation of the Aorta? (LO#1)



What is the most common presentation of children with CoA?

Radiology. Echocardiography with Doppler is the imaging modality of choice, as it can be used to detect the aortic narrowing and estimate the severity.

Chest X-Ray. Chest x-ray is not required for diagnosis, but, in adults with uncorrected CoA, it may reveal “notching” of the inferior surface of the ribs, mostly noted from the third to the eighth ribs (Figure 4).



What is the preferred method of treatment of CoA in older children and adults?

What Is Congenital Pulmonary Valve Stenosis? (LO#1)

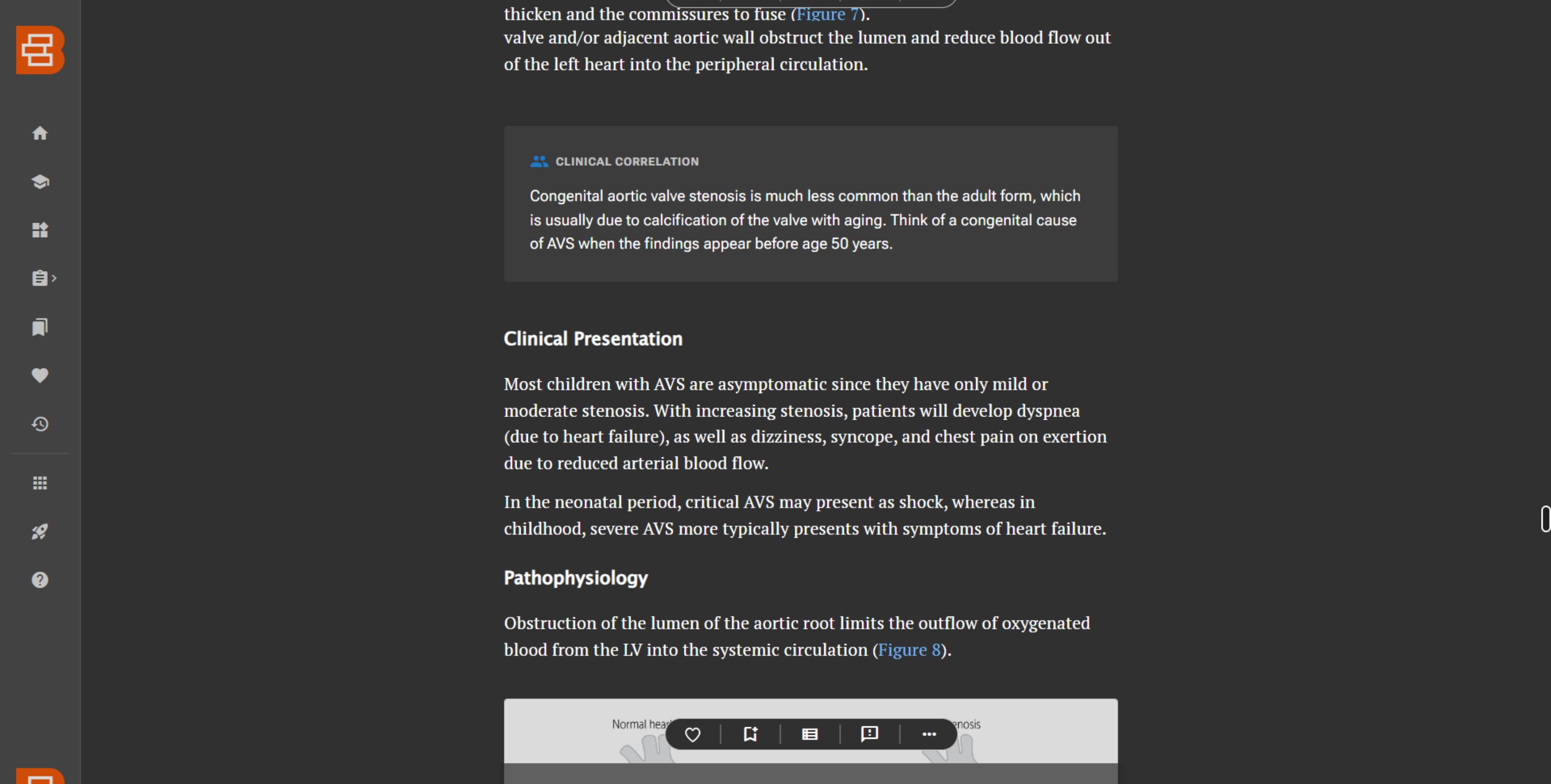
Pulmonary valve stenosis, also called pulmonary stenosis and pulmonary artery stenosis, is the third most common type of CHD, accounting for ~8% of CHD cases. A thickened pulmonary valve narrows the artery’s lumen and limits blood flow into the pulmonary artery and lungs.

Clinical Presentation

The symptoms of pulmonary valve stenosis depend on the degree of narrowing (obstruction). Mild stenosis is typically asymptomatic, and most patients with this defect have mild disease. Severe stenosis, however, causes a drastic decrease in blood flow to the lungs (and therefore in oxygenation), leading to syncope, chest pain, and dyspnea on exertion (all due to hypoxemia).

Pathophysiology

In congenital pulmor pulmonary valve to thicken and the commissures to fuse (Figure 7).
valve and/or adjacent aortic wall obstruct the lumen and reduce blood flow out of the left heart into the peripheral circulation.



CLINICAL CORRELATION

Congenital aortic valve stenosis is much less common than the adult form, which is usually due to calcification of the valve with aging. Think of a congenital cause of AVS when the findings appear before age 50 years.

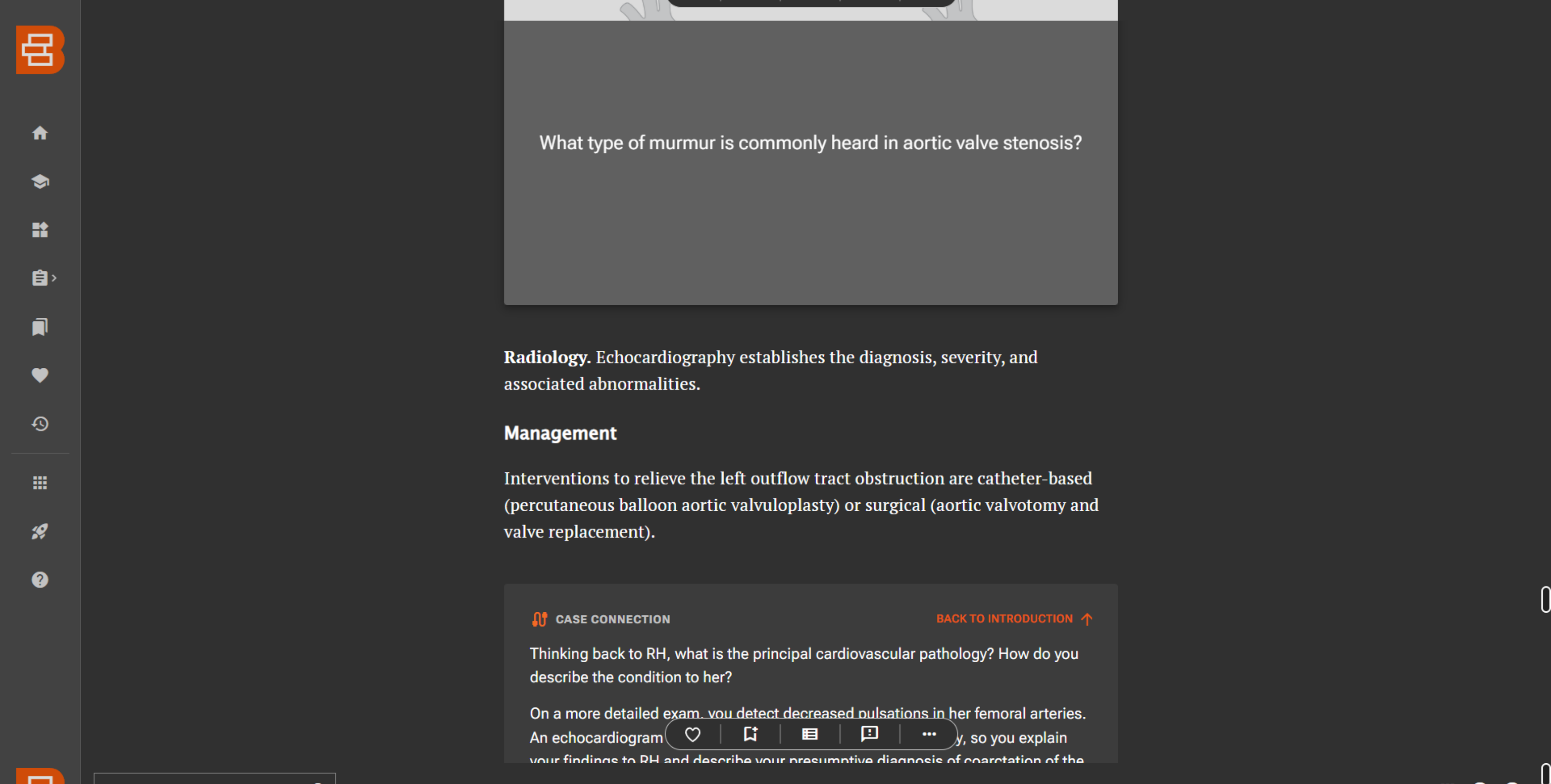
Clinical Presentation

Most children with AVS are asymptomatic since they have only mild or moderate stenosis. With increasing stenosis, patients will develop dyspnea (due to heart failure), as well as dizziness, syncope, and chest pain on exertion due to reduced arterial blood flow.

In the neonatal period, critical AVS may present as shock, whereas in childhood, severe AVS more typically presents with symptoms of heart failure.

Pathophysiology

Obstruction of the lumen of the aortic root limits the outflow of oxygenated blood from the LV into the systemic circulation (Figure 8).



What type of murmur is commonly heard in aortic valve stenosis?

Radiology. Echocardiography establishes the diagnosis, severity, and associated abnormalities.

Management

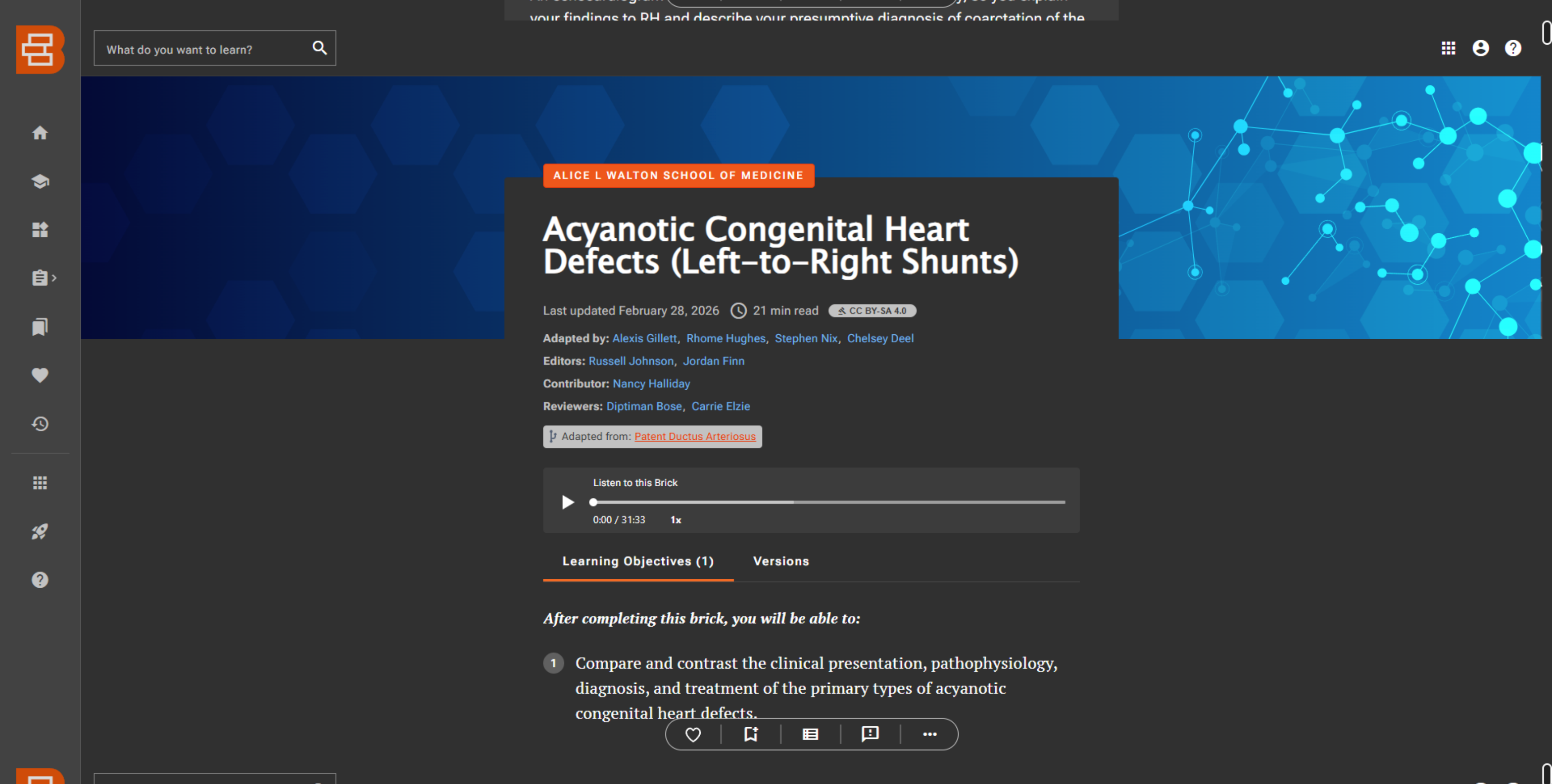
Interventions to relieve the left outflow tract obstruction are catheter-based (percutaneous balloon aortic valvuloplasty) or surgical (aortic valvotomy and valve replacement).

CASE CONNECTION BACK TO INTRODUCTION ↑

Thinking back to RH, what is the principal cardiovascular pathology? How do you describe the condition to her?

On a more detailed exam, you detect decreased pulsations in her femoral arteries. An echocardiogram y, so you explain your findings to RH and describe your presumptive diagnosis of coarctation of the

---



Acyanotic Congenital Heart Defects (Left–to–Right Shunts)

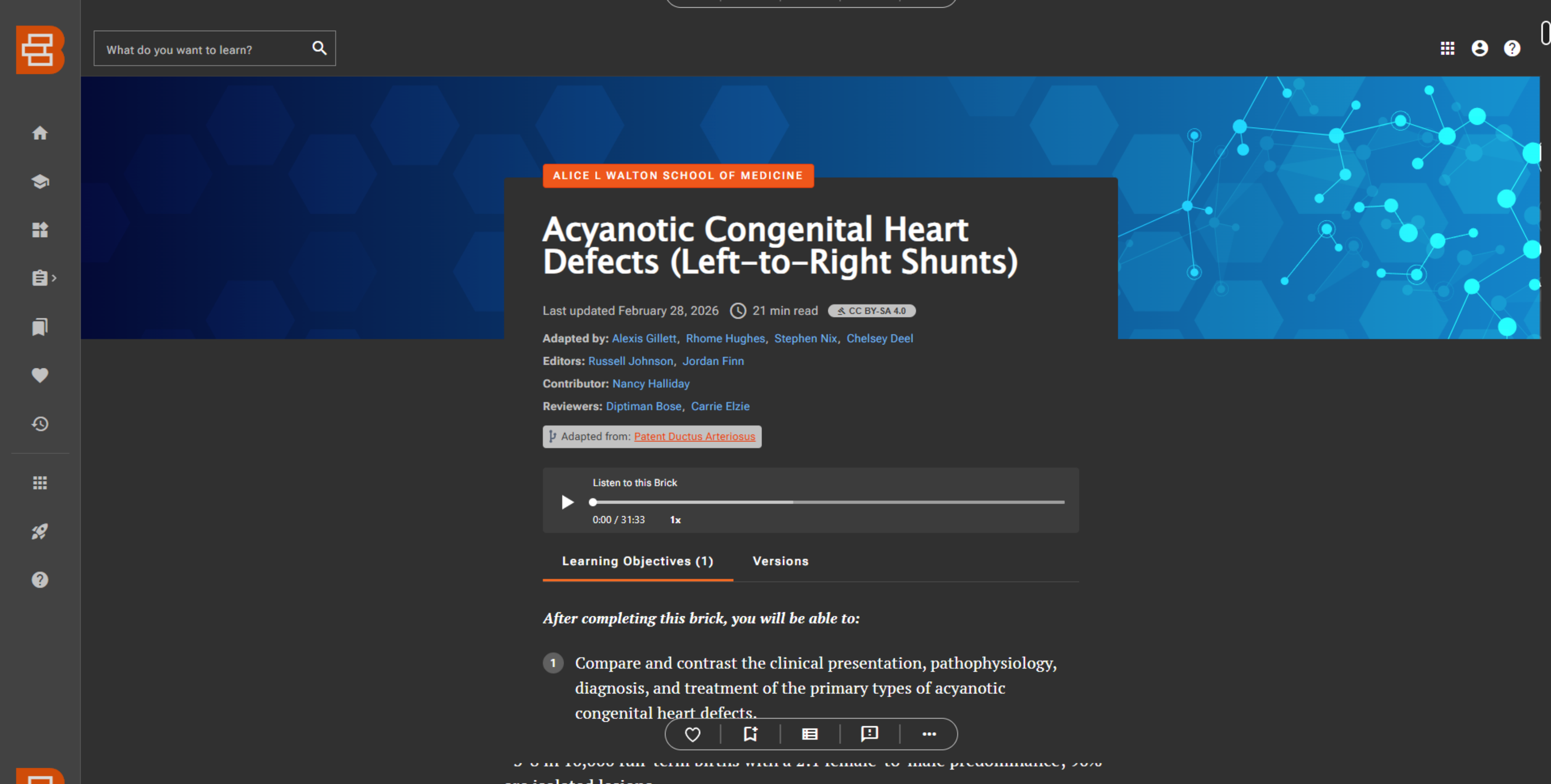
Last updated February 28, 2026 21 min read

Adapted by: Alexis Gillett, Rhome Hughes, Stephen Nix, Chelsey Deel
Editors: Russell Johnson, Jordan Finn
Contributor: Nancy Halliday
Reviewers: Diptiman Bose, Carrie Elzie

Adapted from: Patent Ductus Arteriosus

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of acyanotic congenital heart defects.


Acyanotic Congenital Heart Defects (Left–to–Right Shunts)

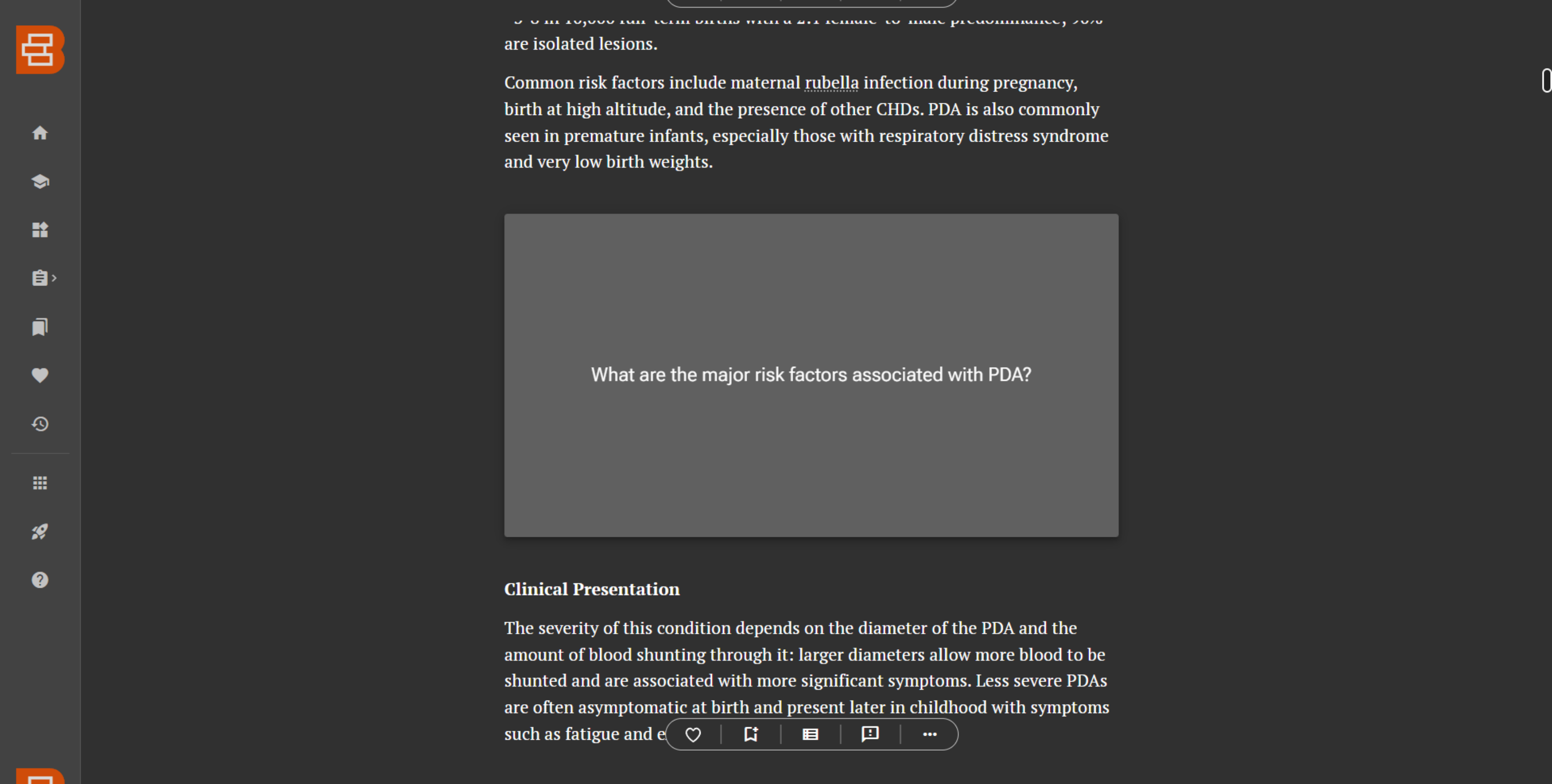
Last updated February 28, 2026 21 min read

Adapted by: Alexis Gillett, Rhome Hughes, Stephen Nix, Chelsey Deel
Editors: Russell Johnson, Jordan Finn
Contributor: Nancy Halliday
Reviewers: Diptiman Bose, Carrie Elzie

Adapted from: Patent Ductus Arteriosus

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of acyanotic congenital heart defects.



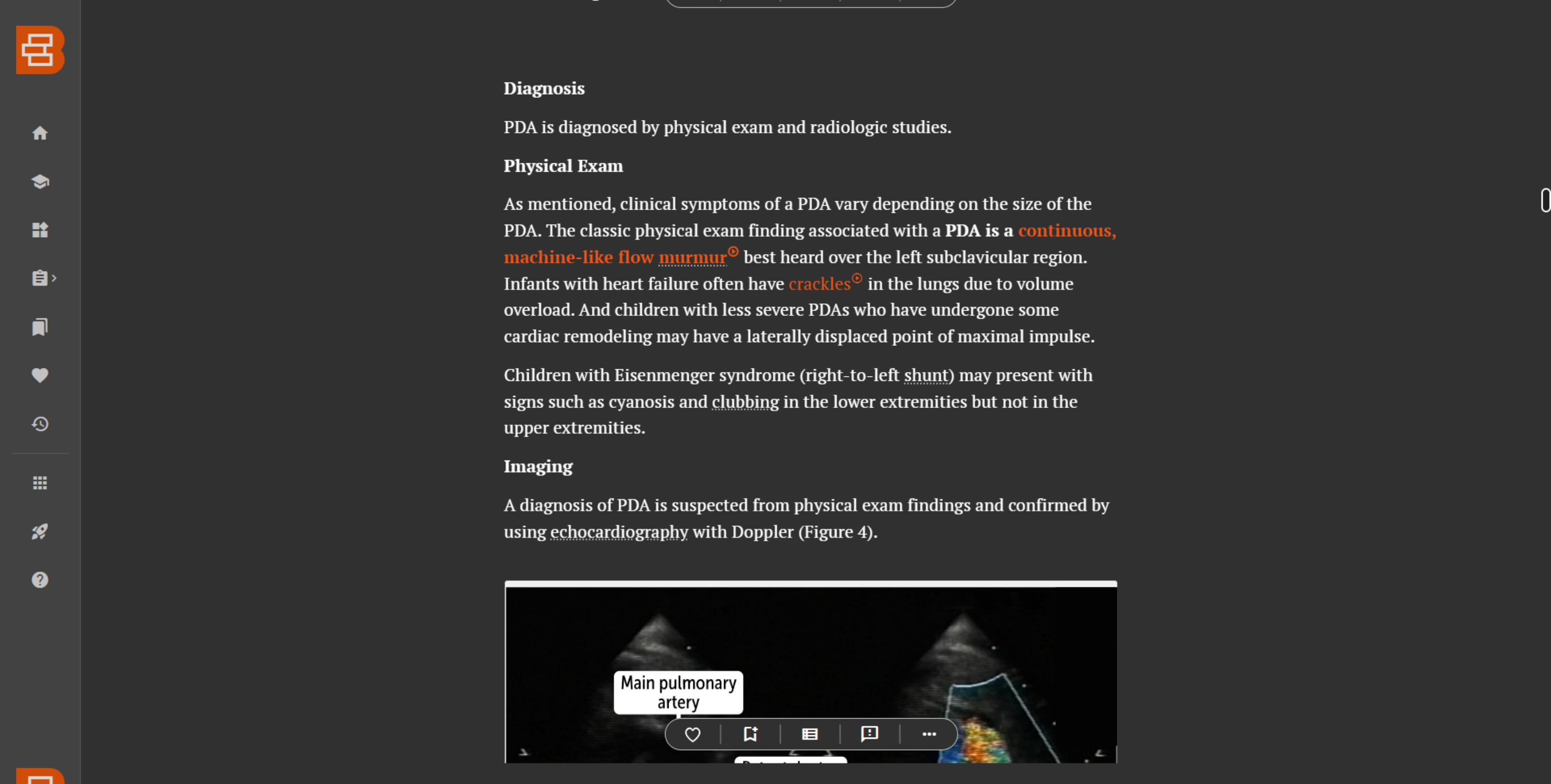
are isolated lesions.

Common risk factors include maternal rubella infection during pregnancy, birth at high altitude, and the presence of other CHDs. PDA is also commonly seen in premature infants, especially those with respiratory distress syndrome and very low birth weights.

What are the major risk factors associated with PDA?

Clinical Presentation

The severity of this condition depends on the diameter of the PDA and the amount of blood shunting through it: larger diameters allow more blood to be shunted and are associated with more significant symptoms. Less severe PDAs are often asymptomatic at birth and present later in childhood with symptoms such as fatigue and e



Diagnosis

PDA is diagnosed by physical exam and radiologic studies.

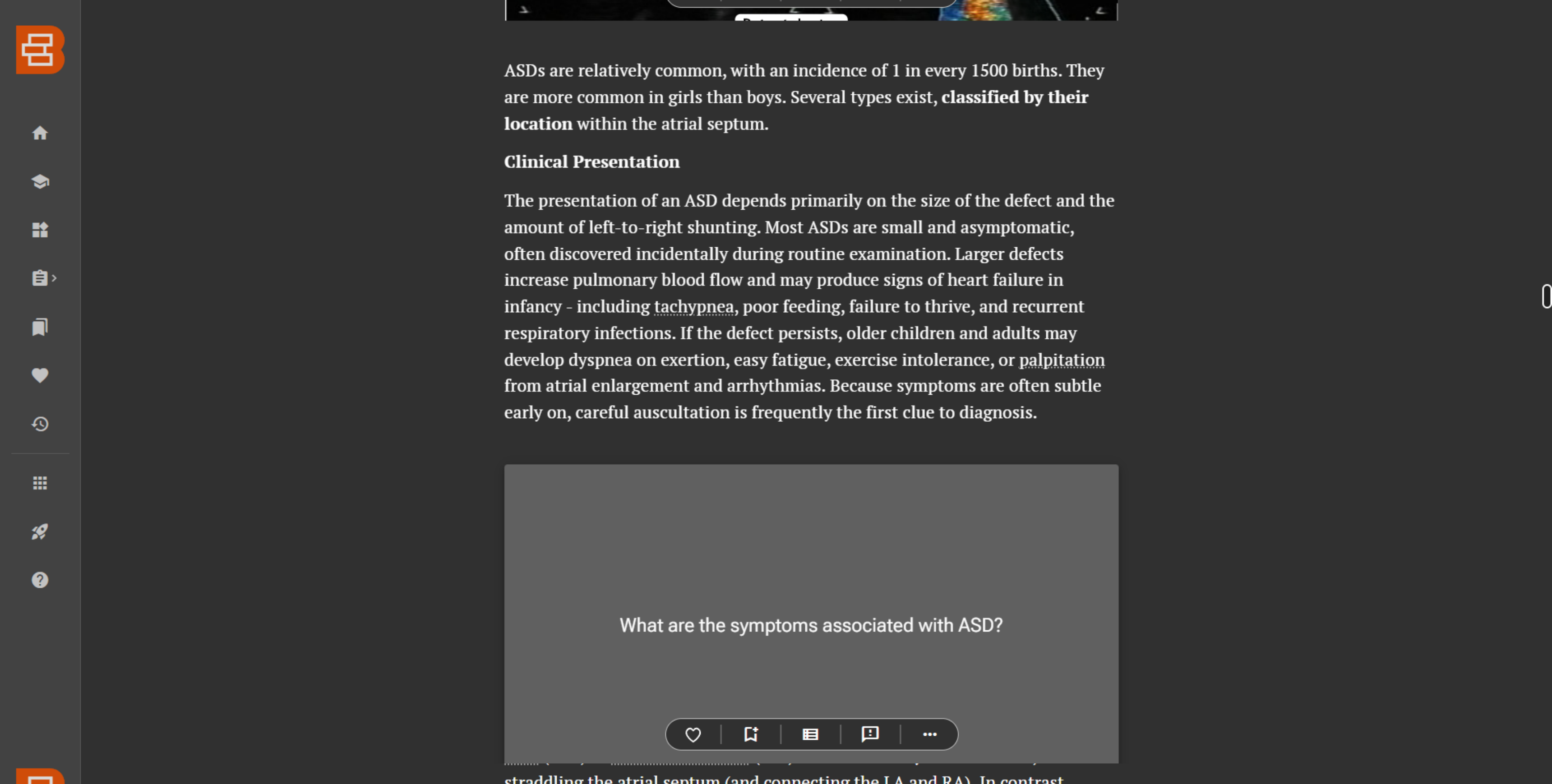
Physical Exam

As mentioned, clinical symptoms of a PDA vary depending on the size of the PDA. The classic physical exam finding associated with a **PDA is a** **continuous, machine-like flow murmur** best heard over the left subclavicular region. Infants with heart failure often have crackles in the lungs due to volume overload. And children with less severe PDAs who have undergone some cardiac remodeling may have a laterally displaced point of maximal impulse.

Children with Eisenmenger syndrome (right-to-left shunt) may present with signs such as cyanosis and clubbing in the lower extremities but not in the upper extremities.

Imaging

A diagnosis of PDA is suspected from physical exam findings and confirmed by using echocardiography with Doppler (Figure 4).

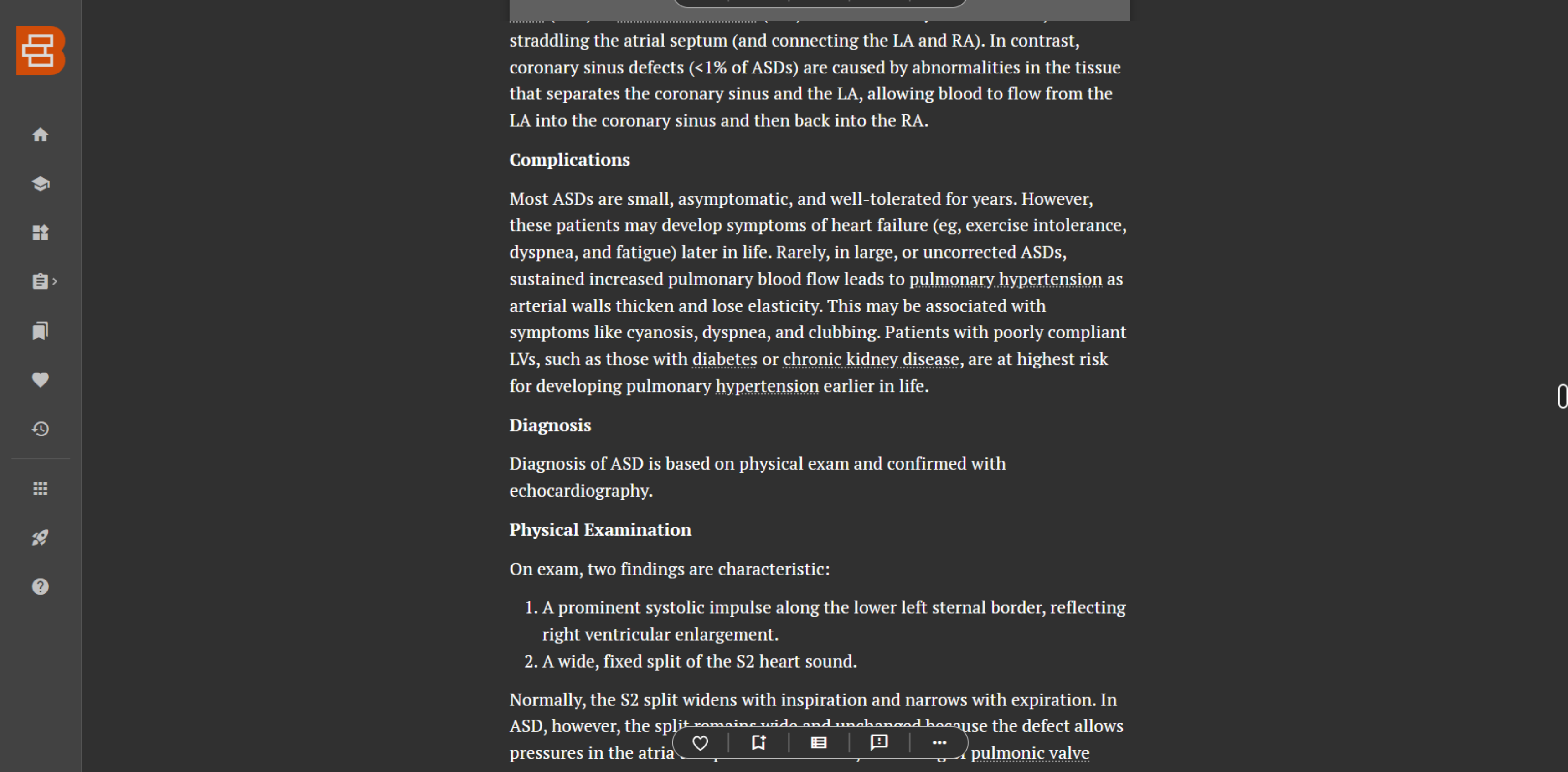


ASDs are relatively common, with an incidence of 1 in every 1500 births. They are more common in girls than boys. Several types exist, **classified by their location** within the atrial septum.

Clinical Presentation

The presentation of an ASD depends primarily on the size of the defect and the amount of left-to-right shunting. Most ASDs are small and asymptomatic, often discovered incidentally during routine examination. Larger defects increase pulmonary blood flow and may produce signs of heart failure in infancy - including tachypnea, poor feeding, failure to thrive, and recurrent respiratory infections. If the defect persists, older children and adults may develop dyspnea on exertion, easy fatigue, exercise intolerance, or palpitation from atrial enlargement and arrhythmias. Because symptoms are often subtle early on, careful auscultation is frequently the first clue to diagnosis.

What are the symptoms associated with ASD?



straddling the atrial septum (and connecting the LA and RA). In contrast, coronary sinus defects (<1% of ASDs) are caused by abnormalities in the tissue that separates the coronary sinus and the LA, allowing blood to flow from the LA into the coronary sinus and then back into the RA.

Complications

Most ASDs are small, asymptomatic, and well-tolerated for years. However, these patients may develop symptoms of heart failure (eg, exercise intolerance, dyspnea, and fatigue) later in life. Rarely, in large, or uncorrected ASDs, sustained increased pulmonary blood flow leads to pulmonary hypertension as arterial walls thicken and lose elasticity. This may be associated with symptoms like cyanosis, dyspnea, and clubbing. Patients with poorly compliant LVs, such as those with diabetes or chronic kidney disease, are at highest risk for developing pulmonary hypertension earlier in life.

Diagnosis

Diagnosis of ASD is based on physical exam and confirmed with echocardiography.

Physical Examination

On exam, two findings are characteristic:

1. A prominent systolic impulse along the lower left sternal border, reflecting right ventricular enlargement.
2. A wide, fixed split of the S2 heart sound.

Normally, the S2 split widens with inspiration and narrows with expiration. In ASD, however, the split remains wide and unchanged because the defect allows pressures in the atria pulmonic valve